

OLS: Ordinary Least Squares

Hypothesis Testing: Linear Regression

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Where We Left Off

Correlation (Pearson's r) tests the relationship between pairs of observations across two variables. It's best used between continuous/interval variables (but hold that thought...)

The “correlation coefficient” indicates how strong the relationship is and the direction it goes in (negative/positive):

- 0.0 to 0.3: weak
- 0.3 to 0.7: moderate
- 0.7 to 1.0: strong

We can test a hypothesis in R if we really want to...

- H_0 : there is no relationship between the variables
- H_1 : there is a relationship between the variables

But we rarely use correlation as a *test* on its own; instead, we use it for a quick glance at association before moving on to other tests.

Let's Clarify Some Stuff

Model: how we describe a system of relationships using mathematical language.

- When we plug variables into a statistical test (e.g. t-test, chi-square test, etc.), we have created a model of the relationship.

Technical interpretation: we have to interpret the technical details of our models (what numbers are important and what do they indicate?).

- E.g.: p-values, number of observations, options (like Cramér's V , direction and size of relationships, “goodness of fit” (which we'll talk about later...))

Substantive interpretation: we also have to interpret the substantive implications of our models using words (what does the whole model suggest?).

- Once we know the technical details, how can we appropriately summarize what this means in real life?
- E.g.: Republicans are twice as likely to vote for this policy as are Democrats.

Regression Analysis

Regression helps us *predict* outcomes (DV) based on inputs (IV).

- “Regression” usually refers to “*linear* regression” (a.k.a., “ordinary least squares regression (OLS)”
- Linear (OLS) regression requires a continuous/interval DV.

With regression, we are testing how different the predicted trend is from what we’d expect if there were no relationship.

- H_0 : the slope of the predictor (IV) is not different from 0 ($\beta_1 = 0$)
- H_1 : the slope of the predictor (IV) is different from 0 ($\beta_1 \neq 0$)

The “linear” part

Regression analysis takes all of the data points between our variables and tries to fit a **line** to them.

$$y = mx + b$$

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Regression line equation:

$$Y_i = \alpha + \beta_i X_i + \epsilon_i$$

α : the Y-intercept

In a regression analysis, **the Y-intercept** will appear as the “Constant” coefficient or the “(Intercept)” coefficient in R.

The intercept is the value of the DV when the IV equals zero.

E.g.: if the α constant is **3.7**, this means that when “Political Knowledge” is equal to zero, on average, “Coolness” will be 3.7.

β : the slope

When we run a regression analysis, the **slope** will appear as the “B” (beta) coefficient.

The β coefficient is the rate of change of the DV as the IV increases.

Importantly, it gives us the direction and the magnitude (“effect size”) of the relationship!

E.g.: if the β coefficient is **5.87**, this means that for every one-unit increase in “Political Knowledge,” there is a 5.87-unit increase in “Coolness.”

ϵ : the error term

The **error (ϵ) term** accounts for all the other random “noise” that can affect our DV outside of the IV.

“Error” doesn’t mean the data is wrong; it means that there is random other stuff that makes certain points deviate from the line.

It won’t show up when we run the regression itself, but it’s crucial to have in the background...

The “ordinary least squares” part

In regression want to minimize the “sum of squared errors”

- In less math-y terms, we want to find the line that minimizes all of those distances (between the points and the line the model predicts).
- We *square* the errors (i.e. errors^2) because it makes them all positive.

The best linear model will have the smallest sum of squared errors.

Interpreting Regression Output

There are three big things to look for in a regression:

1. The significance of the model
2. The relationships between the IV(s) and DV.
3. The “goodness of fit” of the model

The “goodness of fit” is a metric for how well the model predicts the DV.

In regression analysis, we look at the R-Square (or the Adjusted R-Square...)

- In *bivariate* regression, R-square is literally the correlation.
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- 0.7 to 1.0: strong